

MIT

Academy of
Engineering

Department of Computer Engineering

Presentation for TY Major Project-1 (Review 3), Sem I, 2025-26

**Application of Blockchain and Machine Learning in Defence Systems
(Threat Detection / Autonomous Operations / Supply Chain and Data
Management)**

Students Name:

Guide:
Dr. ShitalKumar Jain

- Rohit Lambole
- Ashok Mayekar
- Aditya Patil
- Shravan Mole

Index

- Concept of the project
- Problem Statement and Objectives
- Proposed Block Diagram and its short explanation
- Methodology
- Initial Design / Mathematical Understanding
- Basic Implementation (whatever little possible))
- Selected References

Problem Statement

Our project introduces a Blockchain-based Ammunition Supply Chain System which tackles:

- Manual paperwork → slow approval process
- Lack of transparency in how ammunition moves across hierarchy
- Approvals depend on trust rather than tamper-proof audit logs
- Difficulty in tracking responsibility at each stage
- No unified ledger → discrepancies can go unnoticed
- Potential risk of:

Over-allocation

Unauthorized issue

Mismanagement of critical assets

Motivation & Real Incidents

- Why this project is needed
- Past reports of ammunition pilferage and missing inventory
- Defence audit teams often find discrepancies due to lack of robust tracking
- Traditional centralised databases can be modified or manipulated
- A blockchain-based ledger ensures:
 - Immutability
 - Role-based access
 - Guaranteed accountability

Research Gap

Existing Systems:

Current ammunition management relies on ERP tools, spreadsheets, and centrally controlled approval systems. These solutions record data but cannot guarantee integrity, transparency, or independent verification.

Limitations:

- Records can be altered without strong cryptographic protection
- Audit trails are incomplete or easily manipulated
- Access control relies on a single system admin
- Approvals follow a hierarchical structure but lack verifiable traceability

Research Gap

No existing system provides a tamper-proof, decentralised, cryptographically verifiable ledger for ammunition requests and approvals, where every role's action is securely recorded and independently auditable.

Objective of the Project

Main Objectives

- Build a tamper-proof ledger for ammunition request & approval
- Implement strict role-based access control (RBAC) using smart contracts
- Provide end-to-end transparency
- Ensure zero manual modification of records

Security & Transparency Goals

- Ensure end-to-end traceability of ammunition movement across ranks
- Guarantee zero manual modification of records through blockchain immutability
- Eliminate single-point failures by replacing central authority with decentralised verification
- Provide cryptographic proof for every approval, rejection, and inventory release
- Enable real-time verification of roles and permissions directly from the blockchain
- Reduce fraud, leakage, or unauthorised usage through enforced smart contract rules

Existing vs Proposed System

Existing Manual System

- Relies on paper files and local registers, making tracking slow and inconsistent
- Records can be altered or lost, leading to trust issues and weak accountability
- Approval across ranks is time-consuming and prone to delays
- No centralised visibility of where ammunition is in the workflow
- Auditing is difficult, requiring manual verification of scattered documents

Proposed Blockchain System

- Uses an immutable blockchain ledger, ensuring tamper-free records
- Smart contract–based RBAC automatically enforces permissions
- Enables transparent, multi-level approvals with full traceability
- Auditors can instantly view verified on-chain histories (real-time auditing)
- Records cannot be deleted or modified, ensuring complete integrity

Smart Contract

1. Role Definitions

- COMPANY_LEADER
- REGIMENT_HEAD
- DIVISION_HEAD
- CORPS_HEAD
- COMMAND_HEAD
- INVENTORY_OFFICER
- MANUFACTURE
- ASSEMBLER

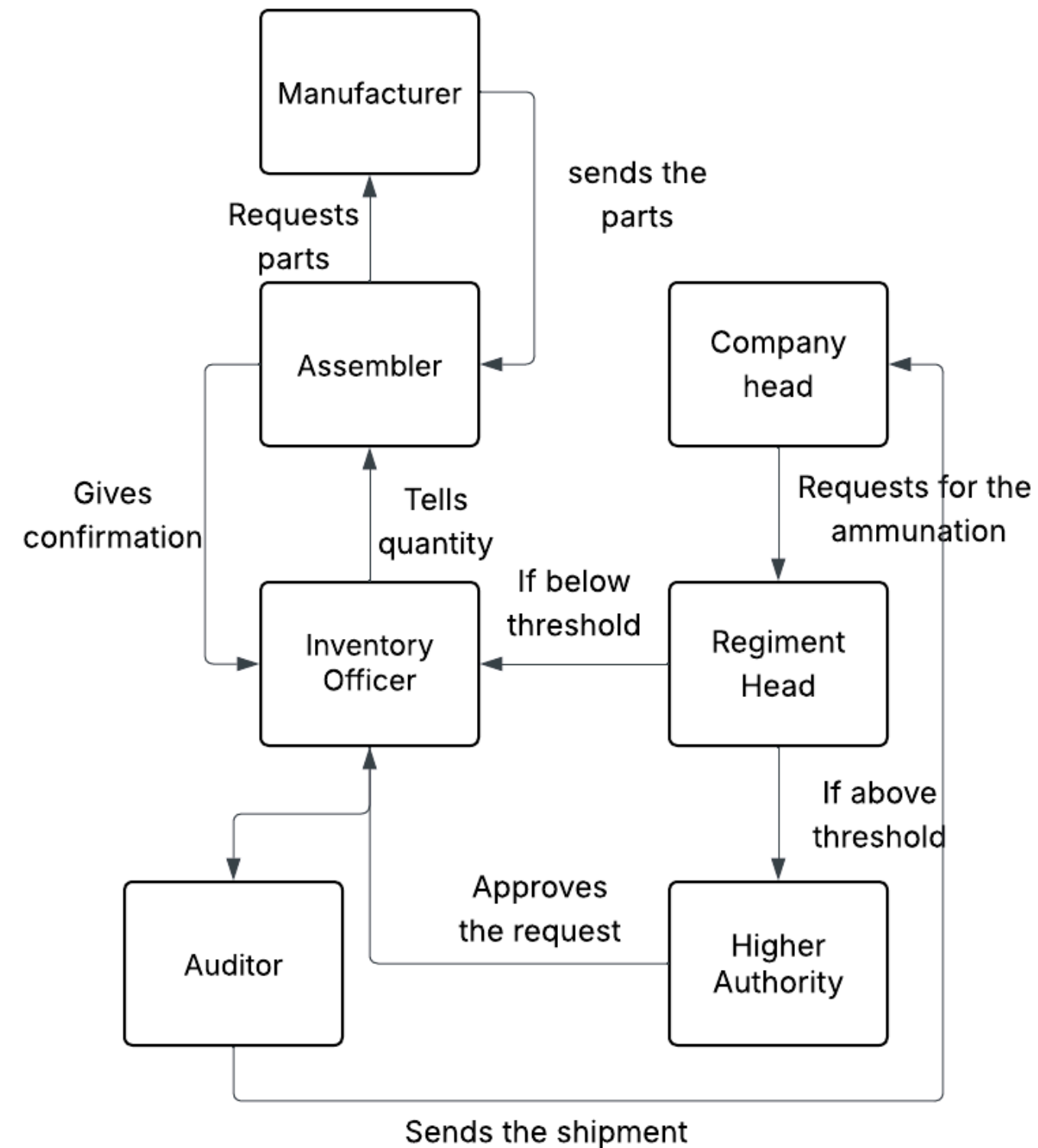
2. Functions

- createRequest()
- approveRequest()
- releaseInventory()
- verifyTransaction()
- grantRole()
- hasRole()

3. Security

- Only defined roles can call specific functions
- Unauthorized calls revert with error (as you saw in tests)
- Blockchain ensures data cannot be altered

Ammunition Request Ledger Workflow



Backend Implementation

Tools Used:

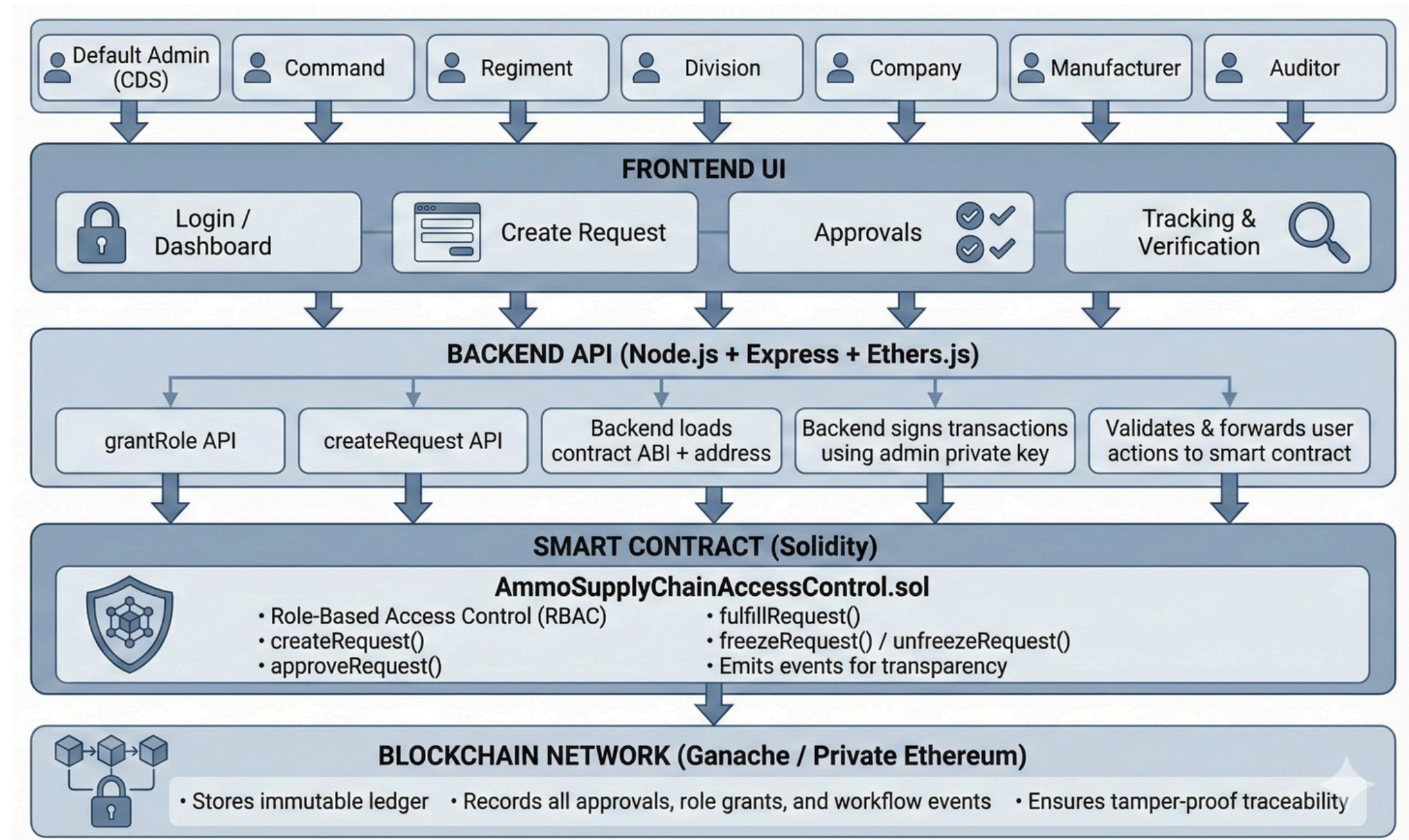
- Node.js
- Express.js
- Ethers.js

API Features:

- Connect to blockchain using RPC
- Load contract using artifact & address
- Role-granting API
- Request creation API

Tested Endpoints:

- POST /grantRole
- GET /checkRole



Smart Contract Deployment

Tools:

- Truffle – Used to compile smart contracts, deploy them, and test blockchain logic on a local development network.
- Ganache – Used to run a private Ethereum blockchain locally with test accounts and private keys.

Deployment Steps:

- 1. `truffle compile` — compiles the Solidity contracts.
- 2. `truffle migrate --reset` — deploys the contracts on the Ganache blockchain.
- 3. Get the contract address — copy the deployed contract address from migration logs.
- 4. Update `backend .env` — set `PRIVATE_KEY`, `CONTRACT_ADDRESS`, so backend connects to the same chain.
- 5. Run backend — `npm run dev` connects backend to deployed contract.

Advantages of Our System

1.Full Transparency:

All authorized roles can see the state changes recorded on the blockchain.

2.Immutable Transaction Log

Once data is written, it cannot be altered or deleted, preventing manipulation of supply-chain records.

3. Reduced Risk of Internal Fraud

Because:i)No one can secretly modify logs

ii)Every action is signed with a private key

iii)Unauthorized access is blocked by smart contract roles

4. Automated Approval Workflow

Smart contracts enforce:

role hierarchy

approval thresholds

permission checks,shipments

Research Papers

Paper Name	Key Outcomes
Blockchain Technology for Military Application (2023)	• Ensures secure and tamper-proof data storage • Improves traceability of ammunition and equipment • Enables faster, automated approvals using smart contracts • Enhances accountability and transparency in operations • Provides resilient and decentralized data management • Supports real-time, trusted data sharing between units • Faces challenges like scalability, integration, and data privacy
Smart Contract-Driven Military Logistics Management System (2023)	• Automates logistics approval and transfer through smart contracts. • Eliminates paper-based processes and human delays. • Increases efficiency in resource allocation and shipment confirmation. • Maintains immutable transaction history for auditing. • Strengthens supply verification and authentication. • Enhances decision-making using verified blockchain data.

Research Papers

Paper Name	Key Outcomes
Subramaniam, C. (2025) Bofors Gate: A Journalist's Pursuit of Truth	• ₹64 crores kickbacks via Swiss accounts show hidden financial trails • Suppressed evidence & lost files since 1997 • Middlemen exploited political proximity • Politicized probes & planted stories → Needs immutable ledger + RBAC + anomaly detection
Ram, N. & Subramaniam, C. (1989) The Bofors Story: Behind the News	• Media blackouts & delayed CBI action • Single-vendor rule flouting & offshore laundering • "Lost evidence" excuses → Requires real-time tamper-proof disclosure & indelible audit trails

Research Papers

Paper Name	Key Outcomes
Pyman, M. et al. (2012) How to Assess Corruption in Defence	• Billions lost yearly to procurement kickbacks • Substandard gear directly endangers soldiers • Political interference & secrecy suppress probes → Justifies RBAC insulation & anomaly freezes
Transparency International (2016) Licence to Bribe? Corruption in High Value Contracts	• AgustaWestland: €51 mn kickbacks via agents • Specification manipulation & unaudited offsets • 74% global defence procurement at high corruption risk → Needs PoA consensus, intermediary logging & locked tender parameters

Thank You!!
Any questions.